Chapter 15, Problem 48P

Problem

Find $f(t)$ using convolution given that:

(a) $F(s) = \dfrac{4}{(s^2 + 2s + 5)^2}$

(b) $F(s) = \dfrac{2s}{(s + 1)(s^2 + 4)}$

Step-by-step solution

Step 1 of 8

(a)

Consider the following function:

$$F(s) = \frac{4}{\left(s^2 + 2s + 5\right)^2}$$

Step 2 of 8

The convolution integral relates the convolution of two signals in the time domain to the inverse of the product of their Laplace transforms:

$$\begin{aligned} \mathrm{L}^{-1}\left[F_1(s)F_2(s)\right] &= f_1(t) * f_2(t) \\ &= \int_0^t f_1(\lambda) f_2(t-\lambda)\, d\lambda \end{aligned}$$

Step 3 of 8

Let us first separate the function into two.

$$\begin{aligned} F(s) &= \frac{4}{\left(s^2 + 2s + 5\right)^2} \\ &= \left(\frac{2}{s^2 + 2s + 5}\right)\left(\frac{2}{s^2 + 2s + 5}\right) \\ &= F_1(s)F_2(s) \qquad \text{since } \left\{ \begin{array}{l} \text{assume:} \\ F_1(s) = \dfrac{2}{s^2 + 2s + 5} \\ F_2(s) = \dfrac{2}{s^2 + 2s + 5} \end{array} \right\} \end{aligned}$$

Step 4 of 8

Find the inverse Laplace transform of the function $F_1(s)$.

$$f_1(t) = \mathrm{L}^{-1}\left[F_1(s)\right]$$
$$= \mathrm{L}^{-1}\left[\frac{2}{s^2+2s+5}\right]$$
$$= \mathrm{L}^{-1}\left[\frac{2}{\left(s^2+2s+1\right)+4}\right]$$
$$= \mathrm{L}^{-1}\left[\frac{2}{(s+1)^2+2^2}\right]$$
$$= e^{-t}\sin 2tu(t) \qquad \text{since } \left\{\mathrm{L}^{-1}\left[\frac{\omega}{(s+a)^2+\omega^2}\right] = e^{-at}\sin\omega t\right\}$$

Similarly for $F_2(s)$

$$f_2(t) = e^{-t}\sin 2tu(t)$$

The product $F_1(s)F_2(s)$ is very complicated; fining the inverse may be tough. In these cases, one must do the convolution in the time domain.

$$F(s)=F_1(s)F_2(s)$$

Apply inverse transform to the above equation.

$$\begin{aligned}
f(t)&=\mathcal{L}^{-1}\left[F(s)\right]\\
&=\mathcal{L}^{-1}\left[F_1(s)F_2(s)\right]\\
&=\int_0^t f_1(\lambda)f_2(t-\lambda)\,d\lambda\\
&=\int_0^t e^{-\lambda}\sin 2\lambda\, e^{-(t-\lambda)}\sin 2(t-\lambda)\,d\lambda\\
&=e^{-t}\int_0^t \sin 2\lambda\sin 2(t-\lambda)\,d\lambda\\
&=\left\{\begin{array}{l}\dfrac{e^{-t}}{2}\displaystyle\int_0^t\cos(2\lambda-2(t-\lambda))d\lambda-\\ \dfrac{e^{-t}}{2}\displaystyle\int_0^t\cos(2\lambda+2(t-\lambda))\,d\lambda\end{array}\right\}\quad \text{since}\left\{\begin{array}{l}\sin x\sin y=\\ \dfrac{1}{2}(\cos(x-y)-\cos(x+y))\end{array}\right\}\\
&=\frac{e^{-t}}{2}\int_0^t\cos(4\lambda-2t)d\lambda-\frac{e^{-t}}{2}\int_0^t\cos(2t)\,d\lambda
\end{aligned}$$

$$=\left\{\begin{array}{l}\left\{\begin{array}{l}\dfrac{e^{-t}}{2}\displaystyle\int_0^t(\cos(4\lambda)\cos(2t)+\sin(4\lambda)\sin(2t))\,d\lambda\\ -\dfrac{e^{-t}\cos(2t)}{2}\displaystyle\int_0^t d\lambda\end{array}\right\}\\ \text{since}\left\{\begin{array}{l}\cos(x-y)=\\ \cos x\cos y+\sin x\sin y\end{array}\right\}\end{array}\right\}$$

Simplified further...

$$\begin{aligned}
f(t)&=\left\{\begin{array}{l}\dfrac{e^{-t}\cos(2t)}{2}\displaystyle\int_0^t\cos(4\lambda)d\lambda+\dfrac{e^{-t}\sin(2t)}{2}\displaystyle\int_0^t\sin(4\lambda)\\ -\dfrac{e^{-t}\cos(2t)}{2}\displaystyle\int_0^t d\lambda\end{array}\right\}\\
&=\left\{\begin{array}{l}\dfrac{e^{-t}\cos(2t)}{2}\left[\dfrac{\sin 4\lambda}{4}\right]_0^t+\dfrac{e^{-t}\sin(2t)}{2}\left[-\cos 4\lambda\right]_0^t\\ -\dfrac{e^{-t}\cos(2t)}{2}\left[\lambda\right]_0^t\end{array}\right\}
\end{aligned}$$

$$=\left\{\begin{array}{l}\dfrac{e^{-t}\cos(2t)}{2}\left[\dfrac{\sin(4t)}{4}-0\right]+\dfrac{e^{-t}\sin(2t)}{2}\left[-\cos 4t+1\right]\\ -\dfrac{e^{-t}\cos(2t)}{2}\left[t-0\right]\end{array}\right\}\quad \text{since }\cos 0^\circ=1$$

Therefore inverse

$$=\frac{e^{-t}\cos(2t)}{2}\left[\frac{\sin(4t)}{4}\right]+\frac{e^{-t}\sin(2t)}{2}\left[1-\cos 4t\right]-\frac{e^{-t}t\cos(2t)}{2}$$

$$=0.125e^{-t}\cos(2t)\sin(4t)+0.5e^{-t}\sin(2t)(1-\cos(4t))-0.5te^{-t}\cos(2t)$$

Laplace transform of $F(s)$ is $f(t)$

$$\boxed{\begin{array}{l}0.125e^{-t}\cos(2t)\sin(4t)u(t)+\\ 0.5e^{-t}\sin(2t)(1-\cos(4t))u(t)-\\ 0.5te^{-t}\cos(2t)u(t)\end{array}}.$$

Step 6 of 8

(b)

Consider the following function:

$$F(s) = \frac{2s}{(s+1)(s^2+4)}$$

Step 7 of 8

The convolution integral relates the convolution of two signals in the time domain to the inverse of the product of their Laplace transforms:

$$\begin{aligned} L^{-1}\left[F_1(s)F_2(s)\right] &= f_1(t) * f_2(t) \\ &= \int_0^t f_1(\lambda) f_2(t-\lambda)\,d\lambda \end{aligned}$$

Let us first separate the function into two.

$$\begin{aligned} F(s) &= \frac{2s}{(s+1)(s^2+4)} \\ &= \left(\frac{s}{s^2+4}\right)\left(\frac{2}{s+1}\right) \\ &= F_1(s)F_2(s) \qquad \text{since } \left\{ \begin{array}{l} \text{assume:} \\ F_1(s) = \dfrac{s}{s^2+4} \\ F_2(s) = \dfrac{2}{s+1} \end{array} \right\} \end{aligned}$$

Step 8 of 8

Find the inverse Laplace transform of the function $F_1(s)$.

$$\begin{aligned} f_1(t) &= \mathrm{L}^{-1}\left[F_1(s)\right] \\ &= \mathrm{L}^{-1}\left[\frac{s}{s^2+4}\right] \\ &= \cos(2t)u(t) \qquad \text{since } \left\{\mathrm{L}^{-1}\left[\frac{s}{s^2+\omega^2}\right] = \cos\omega t\right\} \end{aligned}$$

Similarly for $F_2(s)$

$$\begin{aligned} f_2(t) &= \mathrm{L}^{-1}\left[F_2(s)\right] \\ &= \mathrm{L}^{-1}\left[\frac{2}{s+1}\right] \\ &= 2\mathrm{L}^{-1}\left[\frac{1}{s+1}\right] \\ &= 2e^{-t}u(t) \qquad \text{since } \left\{\mathrm{L}^{-1}\left[\frac{1}{s+a}\right] = e^{-at}\right\} \end{aligned}$$

The product $F_1(s)F_2(s)$ is very complicated; fining the inverse may be tough. In these cases, one must do the convolution in the time domain.

$$F(s) = F_1(s)F_2(s)$$

Apply inverse transform to the above equation.

$$\begin{aligned} f(t) &= \mathrm{L}^{-1}\left[F(s)\right] \\ &= \mathrm{L}^{-1}\left[F_1(s)F_2(s)\right] \\ &= \int_0^t f_1(\lambda)f_2(t-\lambda)\,d\lambda \\ &= \int_0^t \cos(2\lambda)2e^{-(t-\lambda)}\,d\lambda \end{aligned}$$

Simplified further...

$$\begin{aligned} f(t) &= 2e^{-t}\int_0^t e^{\lambda}\cos(2\lambda)\,d\lambda \\ &= 2e^{-t}\left[\frac{e^{\lambda}}{1^2+2^2}(\cos 2\lambda + 2\sin 2\lambda)\right]_0^t \qquad \text{since } \left\{\frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)\right\} \\ &= 2e^{-t}\left[\frac{e^t}{5}\left(\cos(2t)+2\sin(2t)\right) - \frac{e^0}{5}\left(\cos 0^\circ + 2\sin 0^\circ\right)\right] \\ &= 2e^{-t}\left[\frac{e^t}{5}\left(\cos(2t)+2\sin(2t)\right) - \frac{1}{5}\right] \\ &= \frac{2}{5}\cos(2t) + \frac{4}{5}\sin(2t) - \frac{2}{5}e^{-t} \end{aligned}$$

Therefore inverse Laplace transform of $F(s)$ is $f(t)$

$$\boxed{\frac{2}{5}\cos(2t)u(t) + \frac{4}{5}\sin(2t)u(t) - \frac{2}{5}e^{-t}u(t)}.$$